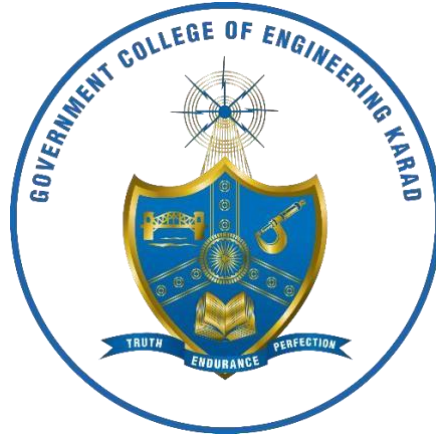


**GOVERNMENT COLLEGE OF ENGINEERING
KARAD
DEPARTMENT OF INFORMATION TECHNOLOGY**



A PROJECT REPORT ON
“To-Do List in Python”

FOR THE AWARD OF

B.Tech in Information Technology
(Second Year, Sem IV)

(2025-26)

UNDER THE GUIDANCE OF

Prof. N. R. SHETTY

SUBMITTED BY

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ABSTRACT

The **To-Do List application** is a simple command-line based project developed using Python to help users manage their daily tasks efficiently. The main objective of this project is to provide an easy-to-use interface for adding, viewing, and deleting tasks. The application allows users to maintain a list of activities, ensuring better organization and time management.

The system uses basic Python concepts such as lists, functions, loops, and file handling. Tasks entered by the user are stored in a text file, which enables data persistence even after the program is closed. This ensures that users can access their tasks at any time without losing information.

Additionally, the program incorporates error handling mechanisms to manage invalid inputs and improve reliability. The project demonstrates the practical implementation of fundamental programming concepts and provides a foundation for developing more advanced task management systems with graphical user interfaces and additional features.

Overall, the To-Do List application is a useful tool for improving productivity and showcases the effective use of Python for solving real-world problems.

Theory:

The To-Do List application is designed to demonstrate the use of basic programming concepts in Python for managing daily tasks. It is a command-line based system that allows users to add, view, and delete tasks in an organized manner. This project highlights how simple programs can be used to solve real-world problems like task management and productivity improvement.

The application primarily uses **lists** as the data structure to store tasks during program execution. Lists in Python are dynamic and allow easy insertion, deletion, and traversal of elements, making them suitable for handling task records.

To ensure that tasks are not lost after the program is closed, the concept of **file handling** is used. Tasks are stored in a text file (tasks.txt) using file operations such as reading (read) and writing (write). This enables data persistence, allowing users to retrieve their saved tasks whenever they run the program again.

The program is structured using **functions**, which help in modularizing the code and improving readability. Each function is responsible for a specific operation such as adding a task, displaying tasks, or removing a task. This approach follows the principle of modular programming.

Control structures such as **loops** and **conditional statements** are used to create a menu-driven interface. The loop ensures continuous execution of the program until the user chooses to exit, while conditional statements handle user choices and direct the flow of execution.

Additionally, the program uses **exception handling (try-except blocks)** to manage errors such as invalid input from the user. This improves the robustness and reliability of the application.

Overall, the project demonstrates how fundamental Python concepts like data structures, file handling, functions, and control flow can be combined to build a practical and user-friendly application.

Program:

To-Do List Application

```
tasks = []
```

```
# Load tasks from file
```

```
def load_tasks():
```

```
    try:
```

```
        with open("tasks.txt", "r") as file:
```

```
            for line in file:
```

```
                tasks.append(line.strip())
```

```
    except FileNotFoundError:
```

```
        pass
```

```

# Save tasks to file
def save_tasks():
    with open("tasks.txt", "w") as file:
        for task in tasks:
            file.write(task + "\n")

# Show tasks
def show_tasks():
    if len(tasks) == 0:
        print("\nNo tasks found.")
    else:
        print("\nYour Tasks:")
        for i, task in enumerate(tasks, start=1):
            print(f'{i}. {task}')

# Add task
def add_task():
    task = input("Enter new task: ")
    tasks.append(task)
    save_tasks()
    print("Task added successfully!")

# Remove task
def remove_task():
    show_tasks()
    try:
        task_no = int(input("Enter task number to remove: "))
        if 1 <= task_no <= len(tasks):
            removed = tasks.pop(task_no - 1)
            save_tasks()
            print(f'Task '{removed}' removed.")
        else:
            print("Invalid task number.")
    except ValueError:
        print("Please enter a valid number.")

# Main program
def main():
    load_tasks()

    while True:
        print("\n===== TO-DO LIST MENU =====")
        print("1. View Tasks")
        print("2. Add Task")
        print("3. Remove Task")

```

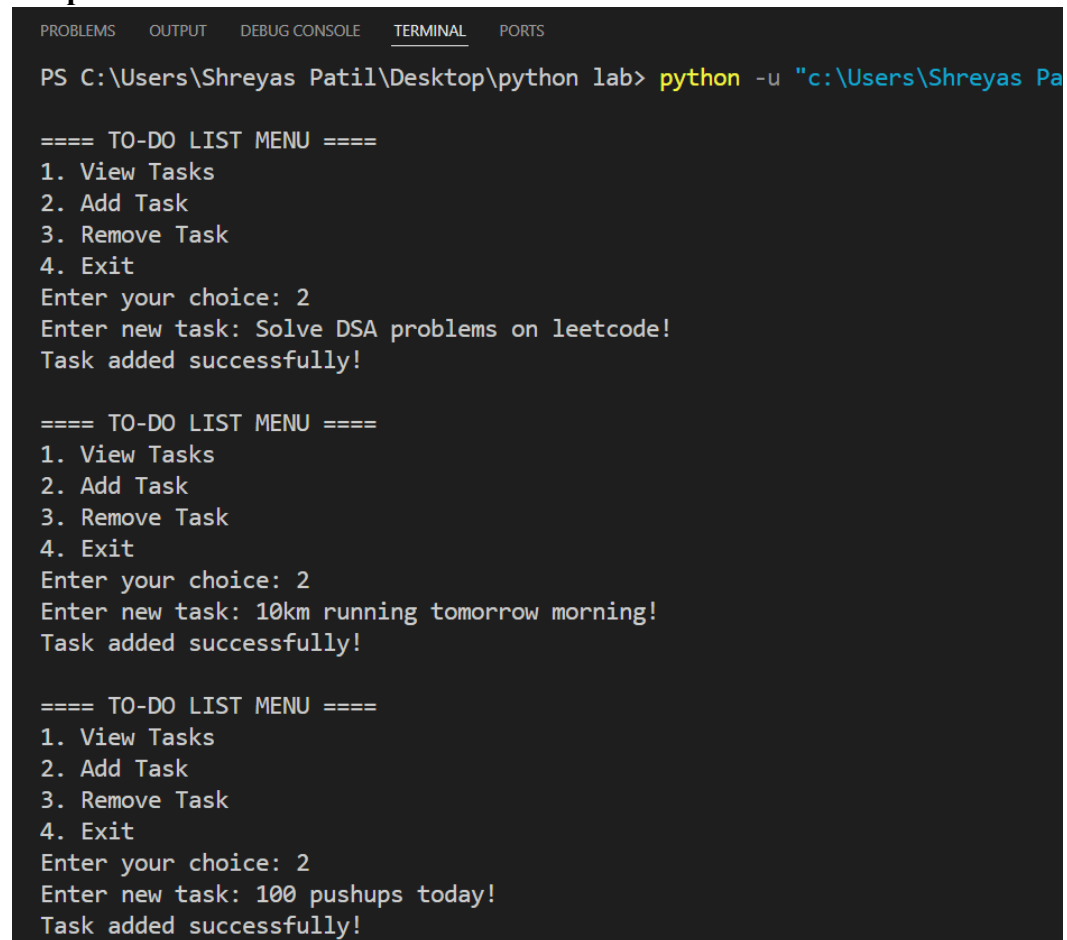
```
print("4. Exit")

choice = input("Enter your choice: ")

if choice == "1":
    show_tasks()
elif choice == "2":
    add_task()
elif choice == "3":
    remove_task()
elif choice == "4":
    print("Exiting... Goodbye!")
    break
else:
    print("Invalid choice, try again.")

# Run program
main()
```

Output:



```
PS C:\Users\Shreyas Patil\Desktop\python lab> python -u "c:\Users\Shreyas Pa

==== TO-DO LIST MENU ====
1. View Tasks
2. Add Task
3. Remove Task
4. Exit
Enter your choice: 2
Enter new task: Solve DSA problems on leetcode!
Task added successfully!

==== TO-DO LIST MENU ====
1. View Tasks
2. Add Task
3. Remove Task
4. Exit
Enter your choice: 2
Enter new task: 10km running tomorrow morning!
Task added successfully!

==== TO-DO LIST MENU ====
1. View Tasks
2. Add Task
3. Remove Task
4. Exit
Enter your choice: 2
Enter new task: 100 pushups today!
Task added successfully!
```

Enter your choice: 1

Your Tasks:

1. Complete JavaScript course today!!
2. complete python projrct report!!
3. Complete DBMS project today!
4. Do 100 pushups!
5. Solve DSA problems on leetcode!
6. 10km running tomorrow morning!
7. 100 pushups today!

==== TO-DO LIST MENU ====

1. View Tasks
2. Add Task
3. Remove Task
4. Exit

Enter your choice: 3

Your Tasks:

1. Complete JavaScript course today!!
2. complete python projrct report!!
3. Complete DBMS project today!
4. Do 100 pushups!
5. Solve DSA problems on leetcode!
6. 10km running tomorrow morning!
7. 100 pushups today!

Enter task number to remove: 1

Task 'Complete JavaScript course today!!' removed.

==== TO-DO LIST MENU ====

1. View Tasks
2. Add Task
3. Remove Task
4. Exit

Enter your choice: 3

Your Tasks:

1. complete python projrct report!!
2. Complete DBMS project today!
3. Do 100 pushups!
4. Solve DSA problems on leetcode!
5. 10km running tomorrow morning!
6. 100 pushups today!

Enter task number to remove: 5

Task '10km running tomorrow morning!' removed.

==== TO-DO LIST MENU ====

1. View Tasks
2. Add Task
3. Remove Task
4. Exit

Enter your choice: 4

Exiting... Goodbye!

PS C:\Users\Shreyas Patil\Desktop\python lab> |

Conclusion:

The To-Do List application developed in Python provides a simple and effective way to manage daily tasks. It allows users to add, view, and delete tasks while storing data using file handling for future use. The project demonstrates key programming concepts such as lists, functions, loops, and exception handling. Overall, it is a practical application that shows how Python can be used to solve everyday problems efficiently.